A Mathematical Framework for Price Oracles

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Abstract

We develop a mathematical framework for price oracles grounded in the economics of arbitrage. Our starting point is the *Consolidated Order Book (COB)*: we define the fair price f_t across n heterogeneous exchanges as the Chebyshev centre of the post-arbitrage-exhaustion spread, and prove it is Lipschitz-stable with constant $\eta/2$ per dollar of depth perturbed. We then model a permissionless *Decentralised Arbitrage Network (DAN)* of capital-constrained bots and prove exponential convergence to the ε-no-arbitrage cone at a rate $\gamma^{\min}(S)$ determined by bot placement and capital alone, with no mid-epoch replenishment. Optimal placement reduces to a p-centre problem admitting a 2-approximation. Under a formal game-theoretic adversary model, we show that any manipulation requires burning capital proportional to the value settled, and derive the unique optimal publication delay Δt^* in closed form, establishing an intrinsic precision–staleness tradeoff.

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1 Consolidated Order Book

The starting point of our framework is a precise notion of what a "fair price" means when assets are traded across multiple exchanges simultaneously, each with its own order book, tick size, and liquidity profile.

Each exchange maintains a *limit order book*: a list of standing buy orders (bids) and sell orders (asks) at various prices. At any instant, a trader can buy the asset at the lowest posted ask, or sell it at the highest posted bid. When the best bid on one exchange exceeds the best ask on another, a *crossed spread* exists: a riskless profit opportunity that any bot can capture by buying on the cheap exchange and selling on the expensive one. These crossed spreads are the signal that the global market has not yet converged to a single price.

1.1 Price Grids and Embedding

Each exchange quotes prices on its own discrete grid determined by its tick size η_i. To compare and consolidate order books across exchanges, we need a common language. We construct the *finest common grid* $\mathcal{G}$, ie. the coarsest grid that contains every exchange's grid as a subgrid. This is an elementary number-theoretic construction, but it is foundational: without it, orders from different exchanges cannot be meaningfully compared.

Definition 1.1. Let $\mathcal{E} = \{1, \ldots, n\}$. Each exchange $i \in \mathcal{E}$ has tick $\eta_i \in \mathbb{Q}_{>0}$ and grid $\mathcal{G}_i = \{k\eta_i : k \in \mathbb{N}_0,\ k \leq K_i\}$. Write $\eta_i = p_i/q_i$ in lowest terms and set

$$D := \operatorname{lcm}(q_1, \ldots, q_n), \qquad d_i := D\eta_i \in \mathbb{N}, \qquad \eta := \frac{\gcd(d_1, \ldots, d_n)}{D} \in \mathbb{Q}_{>0}$$

The *finest common grid* is $\mathcal{G} := \{p_k := k\eta : 0 \leq k \leq K\}$ with $K := \max_i K_i$ and $m := |\mathcal{G}|$.

Lemma 1.2. *$\mathcal{G}_i \subseteq \mathcal{G}$ for every $i \in \mathcal{E}$.*

Proof. Write $k\eta_i = kd_i/D$. Since $g := \gcd(d_1, \ldots, d_n)$ divides d_i, we have $kd_i/D = (kd_i/g) \cdot \eta$, an integer multiple of η in $[0, K\eta]$. □

Definition 1.3. The *snapshot* of exchange i at time t is $\boldsymbol{x_i}^{(t)} = (\boldsymbol{b_i}^{(t)}, \boldsymbol{a_i}^{(t)}) \in \mathbb{R}_+^{2m}$, where $b_{i,k}^{(t)}$ (resp. $a_{i,k}^{(t)}$) is the resting bid (resp. ask) depth at price p_k on exchange i at time t. We set $b_{i,k}^{(t)} = a_{i,k}^{(t)} = 0$ for $p_k \notin \mathcal{G}_i$. The aggregate depth profiles are

$$\boldsymbol{B}^{(t)} := \textstyle\sum_i \boldsymbol{b_i}^{(t)} \in \mathbb{R}_+^m, \qquad \boldsymbol{A}^{(t)} := \textstyle\sum_i \boldsymbol{a_i}^{(t)} \in \mathbb{R}_+^m. \tag{1}$$

We equip $\mathbb{R}_+^{2m}$ with the ℓ^1 norm $\|\boldsymbol{x}\|_1 = \sum_k (|\boldsymbol{b_k}| + |\boldsymbol{a_k}|)$.

1.2 The COB Map and the Fair Price

Given a snapshot of all exchanges, we aggregate the bid and ask depths into two vectors $\boldsymbol{B}, \boldsymbol{A} \in \mathbb{R}_+^m$ over the common grid. If the aggregate book is *crossed*; that is, the best bid exceeds the best ask, a bot can profitably match the two, eliminating both orders and narrowing the spread. The *COB map* (Algorithm 1) iterates this greedy matching until no further profitable crossing remains. The output is a *residual spread* $[p_\beta, p_\alpha]$ with $p_\beta < p_\alpha$, representing the irreducible gap after all free arbitrage has been exhausted.

Definition 1.4. Profiles $(\boldsymbol{B}, \boldsymbol{A})$ satisfy the *NA condition* if

$$\beta(\boldsymbol{B}) := \max\{k : \boldsymbol{B_k} > 0\} < \alpha(\boldsymbol{A}) := \min\{k : \boldsymbol{A_k} > 0\}. \tag{2}$$

The *residual spread* is the interval $[p_{\beta(\boldsymbol{B})}, p_{\alpha(\boldsymbol{A})}]$.

Definition 1.5. Given a NA pair $(\boldsymbol{B}, \boldsymbol{A})$ with residual spread $[p_\beta, p_\alpha]$, define

$$f(\boldsymbol{B}, \boldsymbol{A}) \;:=\; \frac{p_\beta + p_\alpha}{2} \tag{3}$$

This is the *Chebyshev centre* of $[p_\beta, p_\alpha]$: the unique point minimising $\max(f - p_\beta,\, p_\alpha - f)$. It is the unique price at which a buyer and seller both receive non-negative surplus relative to their best available counterparty. Equivalently, the price at which the exchange is equally uninformative about whether it should be bid or offered.

Input: $(\boldsymbol{B}, \boldsymbol{A}) \in \mathbb{R}_+^m \times \mathbb{R}_+^m$
Output: Residual NA pair $(\tilde{\boldsymbol{B}}, \tilde{\boldsymbol{A}})$ and fair price f_t
$(\tilde{\boldsymbol{B}}, \tilde{\boldsymbol{A}}) \leftarrow (\boldsymbol{B}, \boldsymbol{A})$;
$b \leftarrow \beta(\tilde{\boldsymbol{B}})$;
$a \leftarrow \alpha(\tilde{\boldsymbol{A}})$
while $b \geq a$ **do**
  $v \leftarrow \min(\tilde{\boldsymbol{B}}_{\boldsymbol{b}}, \tilde{\boldsymbol{A}}_{\boldsymbol{a}})$
  $\tilde{\boldsymbol{B}}_{\boldsymbol{b}} \leftarrow \tilde{\boldsymbol{B}}_{\boldsymbol{b}} - v$;
  $\tilde{\boldsymbol{A}}_{\boldsymbol{a}} \leftarrow \tilde{\boldsymbol{A}}_{\boldsymbol{a}} - v$
  if $\tilde{\boldsymbol{B}}_{\boldsymbol{b}} = 0$ **then** $b \leftarrow \max\{k < b : \tilde{\boldsymbol{B}}_{\boldsymbol{k}} > 0\}$
  if $\tilde{\boldsymbol{A}}_{\boldsymbol{a}} = 0$ **then** $a \leftarrow \min\{k > a : \tilde{\boldsymbol{A}}_{\boldsymbol{k}} > 0\}$
end
return $(\tilde{\boldsymbol{B}}, \tilde{\boldsymbol{A}})$;
$f_t \leftarrow (p_b + p_a)/2$

Algorithm 1: COB Arbitrage-Exhaustion Map

Theorem 1.6. *Algorithm 1 terminates in at most $2m$ iterations. Its output $(\tilde{\boldsymbol{B}}, \tilde{\boldsymbol{A}})$ is the unique element of $\mathbb{R}_+^m \times \mathbb{R}_+^m$ satisfying:*

(i) the NA condition: $\beta(\tilde{\boldsymbol{B}}) < \alpha(\tilde{\boldsymbol{A}})$;

(ii) mass preservation: $\boldsymbol{B} - \tilde{\boldsymbol{B}} = \boldsymbol{A} - \tilde{\boldsymbol{A}}$;

(iii) componentwise minimality: $\tilde{\boldsymbol{B}} \leq \boldsymbol{B}$ and $\tilde{\boldsymbol{A}} \leq \boldsymbol{A}$.

Moreover, the f_t is a well-defined Borel-measurable function of the snapshot $(\boldsymbol{x}_i^{(t)})_{i \in \mathcal{E}}$.

Proof. Each iteration of the while loop selects a crossed pair (b, a) with $b \geq a$ and removes a positive volume $v = \min(\tilde{\boldsymbol{B}}_b, \tilde{\boldsymbol{A}}_a) > 0$ from both sides. This strictly decreases the total remaining crossed mass

$$\sum_{k : b_k \geq a_k} (\tilde{\boldsymbol{B}}_k + \tilde{\boldsymbol{A}}_k)$$

by $2v > 0$. The number of iterations is at most the total number of level-clearing events, which is bounded by $2m$ (at most m bid levels and m ask levels can be cleared). When the loop terminates, we have $\beta(\tilde{\boldsymbol{B}}) < \alpha(\tilde{\boldsymbol{A}})$, i.e., the NA condition holds. Uniqueness follows because the algorithm performs the greedy maximal matching: any other pair satisfying (i)–(iii) would have to match exactly the same total volume at each crossed price level, implying $(\tilde{\boldsymbol{B}}, \tilde{\boldsymbol{A}}) = (\boldsymbol{B} - M, \boldsymbol{A} - M)$ for the same matching mass vector M.

The map from the raw snapshots $\boldsymbol{x}_i^{(t)}$ to the aggregate $(\boldsymbol{B}, \boldsymbol{A})$, then to the residual $(\tilde{\boldsymbol{B}}, \tilde{\boldsymbol{A}})$, and finally to f_t is composed of continuous and finite-step operations, hence Borel-measurable. $\square$

The most operationally important property of f_t is its sensitivity to market depth perturbations. We ask ourselves the following question: how much can the fair price f_t move if the order books change?

Theorem 1.7. *Let $\boldsymbol{B}, \boldsymbol{A} \in \mathbb{R}_+^m$ and $\boldsymbol{B}', \boldsymbol{A}' \in \mathbb{R}_+^m$ be any two aggregate depth profiles. Let f and f' be the corresponding fair prices defined in Definition 1.5. Then*

$$|f - f'| \;\leq\; \frac{\eta}{2}\big(\big\|\boldsymbol{B} - \boldsymbol{B}'\big\|_1 + \big\|\boldsymbol{A} - \boldsymbol{A}'\big\|_1\big) \tag{4}$$

Proof. Let $\delta\boldsymbol{B} := \boldsymbol{B}' - \boldsymbol{B}$ and $\delta\boldsymbol{A} := \boldsymbol{A}' - \boldsymbol{A}$. Write $\delta\boldsymbol{B} = \delta\boldsymbol{B}^+ - \delta\boldsymbol{B}^-$ where $\delta\boldsymbol{B}^+, \delta\boldsymbol{B}^- \geq 0$. Denote by $\beta = \beta(\boldsymbol{B})$ (and similarly for the rest). The fair price gives

$$|f - f'| = \frac{\eta}{2}\big|(\beta' - \beta) + (\alpha' - \alpha)\big| \leq \frac{\eta}{2}\big(|\beta' - \beta| + |\alpha' - \alpha|\big).$$

To move the best-bid index from β to $\beta' > \beta$, at least one new price level must become strictly positive. In the worst case (most efficient manipulation), advancing the frontier by k ticks requires at least k units of depth mass (in the normalized units where the minimum non-zero depth at any grid point is 1). Therefore,

$$|\beta' - \beta| \;\leq\; \|\delta\boldsymbol{B}\|_1$$

because $\|\delta\boldsymbol{B}\|_1$ is the total dollar volume added or removed on the bid side. The same reasoning applied to the ask side yields $|\alpha' - \alpha| \;\leq\; \|\delta\boldsymbol{A}\|_1$. Combining these inequalities gives

$$|f - f'| \;\leq\; \frac{\eta}{2}\big(\|\delta\boldsymbol{B}\|_1 + \|\delta\boldsymbol{A}\|_1\big) = \frac{\eta}{2}\big(\big\|\boldsymbol{B} - \boldsymbol{B}'\big\|_1 + \big\|\boldsymbol{A} - \boldsymbol{A}'\big\|_1\big).$$

This bound is tight: it is achieved when the entire perturbation is concentrated exactly at (or immediately beyond) the current frontier on one side only. □

Remark 1.8. The ℓ^1 norms $\|\boldsymbol{B} - \boldsymbol{B}'\|_1$ and $\|\boldsymbol{A} - \boldsymbol{A}'\|_1$ measure the total dollar notional of bids and asks that were added, removed, or modified. The constant $\frac{\eta}{2}$ therefore has a clear operational meaning: the fair price moves by *at most half a tick per dollar* of order-book depth perturbed. In real markets, where exchanges impose a minimum order size $\mu > 1$ (in the same units), the actual sensitivity is even smaller, making the oracle strictly more robust than this bound suggests.

2 DAN: Architecture and Convergence

Section 1 showed that the fair price f_t is a deterministic, Lipschitz-stable function of the consolidated book. That analysis was static: it took a snapshot of all exchanges and computed the best attainable price. In practice, order books are continuously perturbed by exogenous order flow, and the consolidated book will generically be *crossed*. The *Decentralised Arbitrage Network* (DAN) is the system of bots whose sole job is to eliminate these crossed spreads, thereby driving the consolidated book towards the no-arbitrage condition and keeping f_t accurate.

A central modelling choice is that we rely *exclusively* on the DAN for convergence. We make no assumption that outside market participants help close arbitrage opportunities, and we further assume that any bot wishing to replenish its inventory from another exchange requires a round-trip time that is *large* relative to a single epoch Δt (see Assumption 2.3). This yields a conservative, self-contained guarantee: the DAN alone must drive the global crossed spread into the target ε-NA cone, even in the worst case where the broader market is inactive and bots operate on fixed, epoch-locked capital. The resulting convergence rate depends only on bot placement, deployed capital, and network latencies.

2.1 Exogenous Order Flow

Assumption 2.1. For each exchange $i \in \mathcal{E}$, exogenous order arrivals form a *marked point process*

$$\mathcal{N}_i \;:=\; \{(T_{i,k}, Z_{i,k})\}_{k\geq 1}$$

where $T_{i,k}$ are arrival times and $Z_{i,k} \in \mathbb{R}_{>0}$ are order sizes (dollar notional). We allow $\mathcal{N}_i$ to be a *Hawkes process* [11] with baseline intensity $\lambda_i^0 > 0$ and self-exciting kernel $\phi_i : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$:

$$\lambda_i(t) \;:=\; \lambda_i^0 + \int_0^t \phi_i(t-s)\, d\mathcal{N}_i(s), \qquad \|\phi_i\|_{L^1} < 1$$

ensuring stationarity. Order sizes $\{Z_{i,k}\}$ are i.i.d. with $\mu_i := \mathbb{E}[Z_{i,1}] < \infty$ and $\mathbb{E}[Z_{i,1}^2] < \infty$. Order flow processes across exchanges are allowed to be *mutually exciting* (correlated), with cross-excitation kernels ϕ_{ij} satisfying the joint stationarity condition of [12]. The process $(\mathcal{N}_i)_{i\in\mathcal{E}}$ is defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions.

Definition 2.2. The *mean global inflation rate* is

$$\Lambda \;:=\; \mathbb{E}\left[\sum_{i\in\mathcal{E}} \lambda_i(t)\,\mu_i\right] \;=\; \sum_{i\in\mathcal{E}} \frac{\lambda_i^0\,\mu_i}{1 - \|\phi_i\|_{L^1}} \tag{5}$$

where the second equality uses the stationary mean intensity of a Hawkes process [11]. This quantity represents the average rate at which exogenous order flow increases the total depth imbalance across the network, thereby tending to create or widen crossed spreads. By Assumption 2.1, $\Lambda < \infty$ is a fixed constant determined by the market's order flow parameters.

Assumption 2.3. Each bot j begins the epoch $[t_0, t_0 + \Delta t]$ with a fixed inventory $q_j(i, i') \geq 0$ of capital deployable on pair (i, i'). The round-trip time required for bot j to replenish its inventory from any external source satisfies

$$T_{\text{replenish}} \;\gg\; \Delta t$$

Consequently, no bot replenishes capital within a single epoch. The capital vector $(q_j(i, i'))_{j,i,i'}$ is therefore $\mathcal{F}_{t_0}$- measurable and fixed throughout $[t_0, t_0 + \Delta t]$.

Remark 2.4. Assumption 2.3 is the natural operational model for a DAN bot: replenishing inventory requires on-chain or cross-exchange transfers, which take minutes to hours, whereas oracle epochs are sub-second to seconds (as established empirically in §3.3). This assumption makes the convergence result *stronger*: we prove convergence even under the worst case where bots cannot be restocked mid-epoch.

2.2 The ε-NA Cone and Precision Floor

Definition 2.5. Fix a time t. The *global residual crossed spread* is

$$\delta_{\max}(t) \;:=\; \max_{\substack{(i,i')\in\mathcal{E}^2\\ i\neq i'}} \Big(p_{\beta(\boldsymbol{b}_i^{(t)})} - p_{\alpha(\boldsymbol{a}_{i'}^{(t)})}\Big)^+ \tag{6}$$

where $\beta(\boldsymbol{b}_i^{(t)}) = \max\{k : b_{i,k}^{(t)} > 0\}$ and $\alpha(\boldsymbol{a}_{i'}^{(t)}) = \min\{k : a_{i',k}^{(t)} > 0\}$. This measures the worst inter-exchange crossing anywhere in the network at time t.

Definition 2.6. For a threshold $\varepsilon > 0$, the *global ε-NA cone* is

$$\mathcal{K}_\varepsilon \;:=\; \big\{(\boldsymbol{B}^{(t)}, \boldsymbol{A}^{(t)}) : \delta_{\max}(t) \le \varepsilon\big\} \tag{7}$$

When the consolidated book lies in $\mathcal{K}_\varepsilon$, no inter-exchange crossing exceeds ε, and by Theorem 1.7 the fair price f_t is within $\varepsilon/2$ of its true value.

Proposition 2.7. *Exact NA ($\varepsilon = 0$) is unachievable. The natural precision floor is*

$$\varepsilon^* \;:=\; \max\Big(\min_{i,i'\in\mathcal{E}}(c_i + c_{i'}),\ \eta\Big) \tag{8}$$

the larger of the minimum round-trip fee and the grid tick. Three structural obstructions enforce this floor:

(i) ***Transaction fees:*** *any bot trading pair (i, i') requires a crossed spread of at least $c_i + c_{i'} > 0$ to break even;*

(ii) ***Execution latency:*** *even a zero-fee bot cannot react instantaneously; exogenous flow may close a crossing before execution arrives;*

(iii) ***Discrete grid:*** *spreads live on $\mathcal{G}$ with tick η; no spread smaller than η is meaningfully resolvable.*

Any target precision ε must satisfy $\varepsilon > \varepsilon^*$; deploying additional capital or reducing latency below what is needed to reach $\mathcal{K}_{\varepsilon^*}$ yields no further accuracy improvement.

2.3 Network Architecture and Bot Placement

Definition 2.8. Let $(\mathcal{F}, d)$ be a finite metric space of *co-location facilities*, where $d : \mathcal{F}\times\mathcal{F} \to \mathbb{R}_{\ge 0}$ is the median round-trip latency between facilities. For exchange $i \in \mathcal{E}$ co-located at facility $\xi_i \in \mathcal{F}$, define

$$\ell(i, f) \;:=\; d(\xi_i, f) \;\ge\; 0$$

as the one-way network latency from facility f to exchange i. We treat $\ell(i, f)$ as the median value; real latencies exhibit small jitter which affects constants but not the qualitative structure of our results.

Definition 2.9. A *placement* is a set $S \subseteq \mathcal{F}$ of bot-server locations with $|S| = N_s$ fixed. The *combined observation latency* of the best-placed bot in S for the pair (i, i') is

$$\ell_S(i, i') \;:=\; \min_{f \in S} \max\big(\ell(i, f), \ell(i', f)\big) \tag{9}$$

This is the minimum total one-way time, over all server locations, for a bot to receive a price update from both i and i' simultaneously.

Definition 2.10. For a pair (i, i'), order the bots in S by increasing combined latency:

$$\ell_{\sigma(1)}(i, i') \;\le\; \ell_{\sigma(2)}(i, i') \;\le\; \cdots \;\le\; \ell_{\sigma(N_s)}(i, i')$$

where $\sigma = \sigma(i, i')$ is the induced permutation on $\{1, \ldots, N_s\}$. Bot $\sigma(r)$ is the r-th bot to observe a crossing on pair (i, i'). Each bot j has a *deployable capital* $q_j(i, i') \ge 0$ (fixed by Assumption 2.3) and a *minimum profitable spread* $c_j := c_i + c_{i'} = \varepsilon^*$ (the round-trip fee, identical for all bots trading the same pair by Definition 8).

Definition 2.11. Given an initial crossing $\Delta > \varepsilon^*$ on pair (i, i'), define the *residual spread* after bot $\sigma(r)$ has acted:

$$\Delta^{(1)} := \Delta, \qquad \Delta^{(r+1)} := \big(\Delta^{(r)} - v_{\sigma(r)}\big)^+, \quad r \ge 1$$

where

$$v_{\sigma(r)} \;:=\; \mathbf{1}\big[\Delta^{(r)} > \varepsilon^*\big] \cdot \min\big(\Delta^{(r)} - \varepsilon^*,\; q_{\sigma(r)}(i, i')\big)$$

is the volume closed by bot $\sigma(r)$: zero if the residual crossing is no longer profitable, otherwise the lesser of the remaining profitable spread and the bot's available capital. The *total volume closed by the chain* given initial crossing Δ is

$$V(i, i', \Delta) \;:=\; \sum_{r=1}^{N_s} v_{\sigma(r)} \;\le\; \Delta - \varepsilon^*$$

With the bot chain mechanics and capital structure in place, we now analyse how the DAN drives the global crossed spread to zero. The key objects are the centred spread $\tilde{\delta}(t)$, which measures the excess above the structural floor ε^*, and the *maximum combined observation latency*

$$\ell^* \;:=\; \max_{(i,i') \in \mathcal{E}^2} \ell_S(i, i')$$

which governs the time between successive bot chain activations and plays the role of the fundamental time unit of the convergence argument. We set $\gamma^{\min}(S) := 1/\ell^*$.

2.4 The Convergence Theorem

Definition 2.12. The *centred maximum crossed spread* is

$$\tilde{\delta}(t) \;:=\; \big(\delta_{\max}(t) - \varepsilon^*\big)^+$$

the excess of the global spread over the structural floor ε^*.

Remark 2.13. Under Assumptions 2.1 and 2.3, $\tilde{\delta}(t)$ is a càdlàg piecewise-constant process. It jumps upward at exogenous arrival times (an order of size $Z_{i,k}$ increases $\delta_{\max}$ by at most $Z_{i,k}$) and jumps downward at bot execution times (bot $\sigma(r)$ reduces the spread on pair (i, i') by exactly $v_{\sigma(r)}$ at time $\ell_{\sigma(r)}(i, i')$ after the crossing appears).

Lemma 2.14. *Fix a time $t \geq 0$ with $\tilde{\delta}(t) > 0$. Let $(i^*, i^{*'})$ be the pair achieving $\delta_{\max}(t)$, and write $\Delta_t := \tilde{\delta}(t) + \varepsilon^*$ for the current crossing size on that pair. Let*

$$R \;:=\; \min\{r \in \{1, \ldots, N_s\} : \textstyle\sum_{s=1}^{r} v_{\sigma(s)} = V(i^*, i^{*'}, \Delta_t)\}$$

be the index of the last active bot, and set $\tau := \ell_{\sigma(R)}(i^*, i^{*'})$ for its combined observation latency. Under Assumption 2.3:

$$\mathbb{E}\big[\tilde{\delta}(t+\tau) \mid \mathcal{F}_t\big] \;\leq\; \big(\tilde{\delta}(t) - V(i^*, i^{*'}, \Delta_t)\big)^+ + \Lambda\,\tau \tag{10}$$

Proof. We track $\tilde{\delta}$ over $[t, t+\tau]$ by accounting separately for the bot chain and exogenous flow. Bot $\sigma(r)$ observes the crossing and executes at time

$$t + \ell_{\sigma(r)}(i^*, i^{*'})$$

for $r = 1, \ldots, R$. These are the only bots active on this crossing: bot $\sigma(r)$ closes $v_{\sigma(r)}$ of profitable spread at its respective time, in strictly increasing order $t+\ell_{\sigma(1)} < \cdots < t+\ell_{\sigma(R)} = t+\tau$. No bot executes after $t+\tau$ on this crossing by definition of R. The total reduction in $\tilde{\delta}$ due to bot executions over $[t, t+\tau]$ is exactly $V := V(i^*, i^{*'}, \Delta_t)$, applied at the sequence of times $t + \ell_{\sigma(1)}, \ldots, t+\tau$.

Let F denote the total dollar notional of exogenous arrivals across all exchanges during $[t, t+\tau)$. Each arriving order of size $Z_{i,k}$ can increase $\delta_{\max}$ by at most $Z_{i,k}$, so:

$$\tilde{\delta}^-(t+\tau) \;\leq\; \tilde{\delta}(t) + F - V'$$

where $\tilde{\delta}^-(t+\tau)$ is the spread just before the final bot execution at $t+\tau$, and V' is the cumulative volume closed by bots $\sigma(1), \ldots, \sigma(R-1)$ prior to $t+\tau$. More carefully: let $\tilde{\delta}_r$ be the spread just after bot $\sigma(r)$'s execution and just before bot $\sigma(r+1)$'s execution. Over each sub-window $(\ell_{\sigma(r)}, \ell_{\sigma(r+1)})$, exogenous flow increases the spread by at most

$$F_r := \sum_{i,k:T_{i,k} \in [t+\ell_{\sigma(r)}, t+\ell_{\sigma(r+1)})} Z_{i,k}$$

The spread evolves as:

$$\begin{aligned}
\tilde{\delta}_0 &= \tilde{\delta}(t), \\
\tilde{\delta}_r &= (\tilde{\delta}_{r-1} - v_{\sigma(r)})^+ + F_{r-1}, \quad r = 1, \ldots, R
\end{aligned}$$

We prove by induction on r that

$$\tilde{\delta}_r \;\leq\; \Big(\tilde{\delta}(t) - \sum_{s=1}^{r} v_{\sigma(s)}\Big)^+ + \sum_{s=0}^{r-1} F_s \tag{11}$$

Base case $r = 0$: $\tilde{\delta}_0 = \tilde{\delta}(t)$ and the right-hand side is $(\tilde{\delta}(t))^+ + 0 = \tilde{\delta}(t)$, so (11) holds with equality. Assume (11) holds for r. Then:

$$\begin{aligned}
\tilde{\delta}_{r+1} &= \big(\tilde{\delta}_r - v_{\sigma(r+1)}\big)^+ + F_r \\
&\leq \left[\Big(\tilde{\delta}(t) - \sum_{s=1}^{r} v_{\sigma(s)}\Big)^+ + \sum_{s=0}^{r-1} F_s - v_{\sigma(r+1)}\right]^+ + F_r \\
&\leq \Big(\tilde{\delta}(t) - \sum_{s=1}^{r} v_{\sigma(s)} - v_{\sigma(r+1)}\Big)^+ + \sum_{s=0}^{r-1} F_s + F_r \\
&= \Big(\tilde{\delta}(t) - \sum_{s=1}^{r+1} v_{\sigma(s)}\Big)^+ + \sum_{s=0}^{r} F_s
\end{aligned}$$

where we used $(a^+ - b)^+ \le (a-b)^+$, which holds for all $a \in \mathbb{R}$ and $b \ge 0$, together with $(x-b)^+ + c \le (x-b)^+ + c$ combined as $(a^+ + c - b)^+ + c' \le (a-b)^+ + c + c'$ for $c, c' \ge 0$ via:

$$(a^+ + c - b)^+ \;\le\; (a + c - b)^+ \;\le\; (a-b)^+ + c$$

Applying (11) at $r = R$ gives

$$\tilde{\delta}(t+\tau) = \tilde{\delta}_R \;\le\; \big(\tilde{\delta}(t) - V(i^*, i^{*'}, \Delta_t)\big)^+ + F,$$

where $F = \sum_{s=0}^{R-1} F_s$ is total exogenous flow over $[t, t+\tau)$. By Assumption 2.1 and the independent-increments property of the Hawkes process over $[t, t+\tau]$:

$$\mathbb{E}[F \mid \mathcal{F}_t] \;=\; \Lambda\,\tau$$

Taking conditional expectations of the bound from Step 2:

$$\mathbb{E}[\tilde{\delta}(t+\tau) \mid \mathcal{F}_t] \;\le\; \big(\tilde{\delta}(t) - V\big)^+ + \Lambda\,\tau$$

□

Lemma 2.14 bounds the expected spread at the end of a single bot-chain window $[t, t+\tau]$ of length $\tau \le \ell^*$. To obtain a global convergence result, we iterate this bound over successive grid times $t_k = k\ell^*$. The key observation is that within each window $[t_k, t_{k+1}]$, two effects operate: the bot chain reduces the spread by $V(i^*, i^{*'}, \Delta_{t_k})$ over $[t_k, t_k + \tau_k]$, and exogenous flow inflates it by at most $\Lambda\ell^*$ over the full window. Under the capital-sufficiency condition (C), the first effect eliminates the entire profitable spread, so the net expected spread at t_{k+1} is controlled purely by the inflation term $\Lambda\ell^*$. The continuous-time bound then follows by bounding the additional inflation within each residual sub-window $[t_k + \tau_k, t_{k+1}]$.

Theorem 2.15. *Let*

$$\ell^* := \max_{(i,i') \in \mathcal{E}^2} \ell_S(i, i')$$

and $Q_{\min} := \min_{(i,i')} Q(i,i')$ *where* $Q(i,i') := \sum_j q_j(i,i')$. *Define the grid times* $t_k := k\ell^*$ *and* $m_k := \mathbb{E}[\tilde{\delta}(t_k)]$. *Under Assumptions 2.1 and 2.3, and under the* capital-sufficiency condition

$$Q_{\min} \;\ge\; \sup_{t \ge 0} \tilde{\delta}(t) \quad \mathbb{P}\text{-}a.s. \tag{C}$$

the following hold.

(i) ***One-step contraction.*** *For all* $k \ge 0$*:*

$$m_{k+1} \;\le\; \frac{\Lambda}{\gamma^{\min}(S)} \tag{12}$$

(ii) ***Continuous-time bound.*** *For all* $t \ge \ell^*$*:*

$$\mathbb{E}[\delta_{\max}(t)] \;\le\; \varepsilon^* + \frac{2\Lambda}{\gamma^{\min}(S)} \tag{13}$$

(iii) ***Design criterion.*** *The oracle reaches* $\mathcal{K}_\varepsilon$ *for all* $t \ge \ell^*$ *if and only if*

$$\gamma^{\min}(S) \;\ge\; \frac{2\Lambda}{\varepsilon - \varepsilon^*} \tag{14}$$

Proof. **(i)** Fix any $k \geq 0$. At time t_k, let $(i^*, i^{*'})$ be the pair achieving $\delta_{\max}(t_k)$ and let τ_k be the arrival time of the last active bot on that pair (as in Lemma 2.14). By definition of $\ell^* = \max_{ii'} \ell_S(i, i')$, we have that

$$\tau_k = \ell_{\sigma(R)}(i^*, i^{*'}) \leq \max_r \ell_{\sigma(r)}(i^*, i^{*'}) \leq \ell^*$$

Under condition (C) gives $Q(i^*, i^{*'}) \geq \tilde{\delta}(t_k)$ a.s., meaning the *total* chain capital suffices. The chain closes the full profitable spread: $V(i^*, i^{*'}, \Delta_{t_k}) = \tilde{\delta}(t_k)$ a.s. Therefore $(\tilde{\delta}(t_k) - V)^+ = 0$ a.s. Applying Lemma 2.14 at time t_k with window $\tau_k \leq \ell^*$:

$$\mathbb{E}[\tilde{\delta}(t_k + \tau_k) \mid \mathcal{F}_{t_k}] \;\leq\; 0 + \Lambda\, \tau_k \;\leq\; \Lambda\, \ell^* \tag{15}$$

Now $t_k + \tau_k \leq t_k + \ell^* = t_{k+1}$. Since $\tilde{\delta}$ is piecewise constant and non-decreasing between bot executions (Remark 2.13), for any $t' \in [t_k + \tau_k, t_{k+1}]$:

$$\tilde{\delta}(t_{k+1}) \;\leq\; \tilde{\delta}(t_k + \tau_k) + F_{[t_k+\tau_k,\, t_{k+1}]} \tag{16}$$

where $F_{[t_k+\tau_k,\, t_{k+1}]}$ is exogenous flow over the residual window $[t_k+\tau_k, t_{k+1}]$ of length $\ell^* - \tau_k \geq 0$. Taking conditional expectations:

$$\mathbb{E}[\tilde{\delta}(t_{k+1}) \mid \mathcal{F}_{t_k}] \;\leq\; \Lambda\, \tau_k + \Lambda(\ell^* - \tau_k) \;=\; \Lambda\, \ell^*$$

Taking unconditional expectations:

$$m_{k+1} \;\leq\; \Lambda\, \ell^* \;=\; \frac{\Lambda}{\gamma^{\min}(S)} \tag{17}$$

This holds for every $k \geq 0$, giving (12).

(ii) Fix $t \geq \ell^*$. Write $t = t_k + s$ for the unique $k \geq 1$ and $s \in [0, \ell^*)$ such that $k\ell^* \leq t < (k+1)\ell^*$. Since $\tilde{\delta}$ is non-decreasing between bot executions, and all bots on the pair achieving $\delta_{\max}(t_k)$ have completed by $t_k + \ell^* = t_{k+1}$:

$$\tilde{\delta}(t) \;\leq\; \tilde{\delta}(t_k) + F_{[t_k,\, t]}$$

where $\mathbb{E}[F_{[t_k,t]} \mid \mathcal{F}_{t_k}] = \Lambda s \leq \Lambda \ell^*$. Taking unconditional expectations:

$$\mathbb{E}[\tilde{\delta}(t)] \;\leq\; m_k + \Lambda\, \ell^* \;\leq\; \Lambda\, \ell^* + \Lambda\, \ell^* \;=\; \frac{2\Lambda}{\gamma^{\min}(S)}$$

Adding ε^* gives (13).

(iii) Requiring $\varepsilon^* + 2\Lambda/\gamma^{\min}(S) \leq \varepsilon$ and rearranging gives (14). □

The design criterion (14) has a clean interpretation. The rate $\gamma^{\min}(S) = 1/\ell^*$ is determined entirely by the worst-case combined observation latency across all exchange pairs: the slower the fastest bot on the hardest-to-cover pair, the lower the convergence rate. The criterion says that the mean order flow rate Λ (the rate at which the market creates new crossed spreads) must not exceed $(\varepsilon - \varepsilon^*)/2$ per latency window ℓ^*. Equivalently, the oracle can tolerate at most

$$(\varepsilon - \varepsilon^*) \cdot \gamma^{\min}(S)/2$$

dollars per second of depth imbalance before the residual spread exceeds ε. More volatile markets require either tighter bot placement (smaller ℓ^*) or a looser precision target ε. Separately, capital $Q_{\min}$ must be large enough to cover the spread at all times: since the stationary spread concentrates around $\Lambda/\gamma^{\min}(S)$, a practical requirement is

$$Q_{\min} \gtrsim \Lambda/\gamma^{\min}(S)$$

i.e. each bot chain must hold at least one latency window worth of market flow in deployable inventory.

2.5 Optimal Bot Placement

Problem 2.16. Given network $(\mathcal{F}, d)$, exchanges $\mathcal{E}$, capital allocations (q_j), and mean flow rate Λ, find

$$S^* := \underset{S \subseteq \mathcal{F},\ |S| = N_s}{\arg\max}\ \gamma^{\min}(S)$$

Theorem 2.17. *Assume capital allocations $(q_j(i, i'))$ are fixed and equal across all bots for each pair: $q_j(i, i') = Q/N_s$ for all j. Then Problem 2.16 is equivalent to the* weighted p-centre problem*:*

$$S^* = \underset{S \subseteq \mathcal{F},\ |S| = N_s}{\arg\min}\ \max_{(i,i') \in \mathcal{E}^2} \ell_S(i, i') \tag{18}$$

The p-centre problem on a finite metric space is NP-hard in general [8] but admits a 2-approximation via the Gonzalez greedy algorithm [9], which runs in time $O(|\mathcal{E}|^2 \cdot |\mathcal{F}|)$.

3 Manipulation Security and Optimal Publication Delay

The DAN plays a dual role: its bots are simultaneously *arbitrageurs* that enforce price accuracy and *implicit data reporters* whose execution traces constitute a ground-truth audit of every exchange feed. Any exchange that falsifies its feed thereby creates a *phantom arbitrage*: a crossed spread backed by no genuine liquidity. When a DAN bot executes against this phantom wall, the ensuing slippage is measurable and irrefutable. We build on this observation to give a comprehensive security analysis.

Section 3.1 sets up a formal adversary model in a continuous-time game. Section **??** derives an exact manipulation cost function and a tight lower bound on the capital an attacker must burn to keep a crossing alive. Section **??** closes the gap left by the Markov bound with a sharp tail estimate under the Ornstein–Uhlenbeck approximation. Section **??** derives the optimal publication delay Δt^* in closed form and establishes the fundamental precision–staleness tradeoff. Section **??** formally compares the DAN oracle against TWAP and committee oracles under the same security definition.

3.1 Formal Adversary Model

We model oracle security as a two-player zero-sum game between a *protocol* (the DAN) and an *adversary* $\mathcal{A}$. Time is continuous; the oracle publishes at equally-spaced *epochs* $t_0 < t_0 + \Delta t < t_0 + 2\Delta t < \cdots$. We analyse a single epoch $[t_0, t_0 + \Delta t]$ without loss of generality by stationarity of the market model.

Definition 3.1. At every time t, the *public information* is the filtration

$$\mathcal{F}_t \;:=\; \sigma\Big(\big(\boldsymbol{x}_i^{(s)}\big)_{i\in\mathcal{E},\, s\le t},\; \big(S,\, \{q_j\}_{j=1}^N,\, \{\ell(i,f)\}_{i,f}\big)\Big)$$

i.e. the entire history of consolidated order books together with the *publicly announced* bot placement, capital allocations, and latency map. The adversary is given $\mathcal{F}_{t_0}$: full knowledge of the current book and all DAN parameters. Exogenous order flow $\big(\lambda_i(s)\big)_{s>t_0}$ is *not* observed beyond its distributional parameters (λ_i, μ_i).

Definition 3.2. A $(\kappa, \Delta t)$-*admissible adversary strategy* is a $\mathcal{F}_{t_0}$-measurable random variable

$$\mathcal{A} \;=\; \big(\{(p_k^{(j)}, v_k^{(j)}, s_k^{(j)})\}_{j=1}^{J},\; \{(p_k^{(j)}, \tau_k^{(j)})\}_{j=1}^{J'}\big)$$

specifying a finite set of *order placements* $(p_k^{(j)}, v_k^{(j)}, s_k^{(j)}) \in \mathcal{G} \times \mathbb{R}_{>0} \times \{\text{bid}, \text{ask}\}$ and a finite set of *cancellations* at times $\tau_k^{(j)} \in [t_0, t_0 + \Delta t)$, subject to the following constraints:

(i) *Capital budget:* $\displaystyle\sum_j p_k^{(j)} \cdot v_k^{(j)} \;\le\; \kappa$;

(ii) *Epoch lock:* all placements are submitted at t_0; cancellations may occur at any time $\tau_k^{(j)} \in (t_0, t_0 + \Delta t)$, but no new orders may be placed after t_0;

(iii) *No self-crossing:* the adversary's own orders do not cross each other.

Remark 3.3. The epoch lock (constraint (ii)) is the model-level counterpart of the oracle's atomic snapshot: the price is computed from the book at the epoch boundary, and the adversary cannot observe mid-epoch bot reactions to adapt their orders. This is the *strongest reasonable adversary* consistent with an atomic publication: allowing mid-epoch order modification would give the adversary oracle-like information about bot behaviour, which is unrealistic.

Definition 3.4. Fix a snapshot time $t_0 + \Delta t$. Let $(\boldsymbol{B}^{(t_0+\Delta t)}, \boldsymbol{A}^{(t_0+\Delta t)})$ denote the raw order book at that time, including any adversarial orders that have not yet been executed or cancelled. Let $(\boldsymbol{B}^*, \boldsymbol{A}^*)$ denote the order book *after* all profitable arbitrage against adversarial orders has actually been executed by the DAN during $(t_0, t_0 + \Delta t)$.

- The *true fair price* is $f^* := f(\tilde{\boldsymbol{B}}^*, \tilde{\boldsymbol{A}}^*)$, the COB output applied to the real post-execution book.
- The *published price* is $\hat{f} := f(\tilde{\boldsymbol{B}}^{(t_0+\Delta t)}, \tilde{\boldsymbol{A}}^{(t_0+\Delta t)})$, the COB output applied to the raw snapshot book.

The difference $\hat{f} - f^*$ is the *oracle deviation*: the amount by which the adversary's uncancelled, unexecuted phantom wall skews the published price relative to the true post-arbitrage price. By Theorem 1.7, this deviation satisfies

$$|\hat{f} - f^*| \;\leq\; \frac{\eta}{2}\big(\|\boldsymbol{B}^{(t_0+\Delta t)} - \boldsymbol{B}^*\|_1 + \|\boldsymbol{A}^{(t_0+\Delta t)} - \boldsymbol{A}^*\|_1\big)$$

where the ℓ^1 norms measure the total notional of adversarial orders that were neither executed nor cancelled by the snapshot time.

Remark 3.5. The COB algorithm simulates arbitrage exhaustion arithmetically on the snapshot: no actual trades occur. An adversary who posts a limit-order wall inside the spread creates a crossing in the simulation: the COB matches it on paper and reports a shifted fair price. The wall is *phantom* in the sense that the reported crossing has not been resolved by real execution. The attack succeeds if the wall is still present at snapshot time: either because the DAN did not execute against it in time, or because execution is not possible (e.g. the adversary posted the wall on an exchange where DAN bots have no capital). The adversary can attempt to cancel the wall immediately after the snapshot, recovering the unexecuted notional at no cost.

Definition 3.6. The *manipulation event* is

$$\mathcal{M}(\varepsilon, \Delta t) \;:=\; \big\{|\hat{f} - f^*| \;\geq\; \varepsilon/2\big\} \tag{19}$$

i.e. the published price deviates from the true post-execution fair price by at least $\varepsilon/2$. On this event, any smart contract settling at $\hat{f}$ misprices by at least $\varepsilon/2$. Note that $\mathcal{M}(\varepsilon, \Delta t)$ depends on both the adversary's strategy (which determines the phantom wall) and the DAN's execution (which determines how much of the wall was consumed before the snapshot).

Definition 3.7. Given the current aggregate depth profiles $(\boldsymbol{B}, \boldsymbol{A})$, the *sweep cost* $C(\boldsymbol{A}, \varepsilon)$ is the minimum dollar notional of ask depth the adversary must consume (by placing buy orders that match against it) to shift the best ask upward by ε, thereby moving $\hat{f}$ upward by $\varepsilon/2$:

$$C(\boldsymbol{A}, \varepsilon) \;:=\; \sum_{j=0}^{\varepsilon/\eta - 1} A_{\alpha+j} \cdot p_{\alpha+j} \tag{20}$$

where $A_{\alpha+j}$ is the aggregate ask depth at price level $p_{\alpha+j} = p_\alpha + j\eta$. The symmetric definition applies for a downward skew using bid depth $\boldsymbol{B}$. By Theorem 1.7, the sweep cost is bounded below by

$$C(\boldsymbol{A}, \varepsilon) \;\geq\; \frac{\varepsilon}{\eta} \tag{21}$$

with equality when each of the ε/η consumed levels has unit depth. In a thick book with large $A_{\alpha+j}$, the sweep cost is much larger than this floor.

Definition 3.8. The *effective bot capital* available to execute against an adversarial wall on pair (i, i') within a single epoch $[t_0, t_0 + \Delta t]$ is

$$Q(i, i') \;:=\; \sum_{j=1}^{N_s} q_j(i, i') \tag{22}$$

the total capital deployed by all bots on that pair, fixed at t_0 by Assumption 2.3. The *expected wall loss* to the adversary is

$$L(\mathcal{A}) \;:=\; \varepsilon \cdot \min\big(D(\mathcal{A}),\; Q(i^*, i^{*'})\big) \tag{23}$$

where $D(\mathcal{A})$ is the total depth of the adversary's wall on the bottleneck pair $(i^*, i^{*'})$ and ε is the price skew per unit executed.

Lemma 3.9. *The manipulation event $\mathcal{M}(\varepsilon, \Delta t)$ occurs if and only if the adversary's wall on the bottleneck pair has residual depth*

$$D^{\text{res}}(\mathcal{A}, \Delta t) \;:=\; \big(D(\mathcal{A}) - Q(i^*, i^{*'})\big)^+ \;\geq\; \frac{\varepsilon}{\eta} \tag{24}$$

at the snapshot time $t_0 + \Delta t$, i.e. the unexecuted wall depth is large enough that the COB simulation reports a price deviation of at least $\varepsilon/2$.

Proof. By Definition 3.4, the oracle deviation $|\hat{f} - f^*|$ equals the Lipschitz bound applied to the difference between the raw snapshot book and the post-execution book. The difference is exactly the residual wall: the D^{res} units of depth that were neither executed by bots nor cancelled. By Theorem 1.7:

$$|\hat{f} - f^*| \;=\; \frac{\eta}{2} \cdot D^{\text{res}} \cdot \frac{1}{\text{depth units per dollar}}$$

For $|\hat{f} - f^*| \geq \varepsilon/2$ we need $D^{\text{res}} \geq \varepsilon/\eta$ (in the normalised units where the minimum depth per level is 1), giving (24). The bot chain consumes at most $Q(i^*, i^{*'})$ of the wall, so $D^{\text{res}} = (D(\mathcal{A}) - Q(i^*, i^{*'}))^+$. □

Remark 3.10. Lemma 3.9 reveals the precise race condition. The adversary posts a wall s.t.

$$D(\mathcal{A}) > Q(i^*, i^{*'}) + \varepsilon/\eta$$

i.e. large enough that even after bots consume $Q(i^*, i^{*'})$, the residual $D^{\text{res}} \geq \varepsilon/\eta$ remains to skew the oracle. The cost of the attack is:

- *Guaranteed loss:* $L(\mathcal{A}) = \varepsilon \cdot Q(i^*, i^{*'})$, the notional sniped by bots regardless of whether the attack succeeds.
- *Wall cost:* the sweep cost $C(\boldsymbol{A}, \varepsilon)$ to move the best ask by ε. The portion $D^{\text{res}} \cdot p^*$ can be cancelled after the snapshot at zero additional cost.

Definition 3.11. The adversary's *expected net payoff* is

$$\mathbb{E}[\Pi(\mathcal{A})] \;:=\; \frac{\varepsilon}{2} \cdot \text{TVS} \cdot \mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)] \;-\; C(\boldsymbol{A}, \varepsilon) \;-\; L(\mathcal{A}) \tag{25}$$

where:

- $\frac{\varepsilon}{2} \cdot \text{TVS}$ is the gross gain from settling TVS of contracts at a price $\varepsilon/2$ away from the true fair price;
- $C(\boldsymbol{A}, \varepsilon)$ is the sweep cost (Definition 3.7);
- $L(\mathcal{A}) = \varepsilon \cdot Q(i^*, i^{*'})$ is the guaranteed wall loss to DAN bots (Definition 3.8).

3.2 The security game

Definition 3.12. The *oracle security game* $\mathsf{Game}(\varepsilon, \Delta t, \kappa)$ proceeds as follows.

1. The protocol announces $(S, \{q_j\}, \{\ell(i, f)\}, \Delta t)$. The book $(\boldsymbol{B}^{(t_0)}, \boldsymbol{A}^{(t_0)})$ is observed by all parties.
2. The adversary submits a $(\kappa, \Delta t)$-admissible strategy $\mathcal{A}$ (Definition 3.2). Orders are applied to the book at t_0.
3. Exogenous order flow evolves over $(t_0, t_0 + \Delta t)$ under Assumption 2.1. DAN bots execute against any profitable crossing they observe, subject to their capital and latency constraints and Assumption 2.3.
4. At $t_0 + \Delta t$, the oracle publishes $\hat{f} = f(\tilde{\boldsymbol{B}}^{(t_0+\Delta t)}, \tilde{\boldsymbol{A}}^{(t_0+\Delta t)})$.
5. Payoff $\mathbb{E}[\Pi(\mathcal{A})]$ is realised (Definition 3.11).

The oracle is $(\varepsilon, \kappa, \Delta t)$-*secure* if $\mathbb{E}[\Pi(\mathcal{A})] \leq 0$ for every $(\kappa, \Delta t)$-admissible $\mathcal{A}$.

Proposition 3.13. *In the security game* $\mathsf{Game}(\varepsilon, \Delta t, \kappa)$*, the adversary's optimal wall depth on the bottleneck pair is*

$$D^* \;=\; Q(i^*, i^{*'}) + \frac{\varepsilon}{\eta} \tag{26}$$

the minimum depth for which the manipulation event $\mathcal{M}(\varepsilon, \Delta t)$ *occurs with positive probability. Any* $D < D^*$ *guarantees* $\mathcal{M}(\varepsilon, \Delta t)$ *does not occur (Lemma 3.9). Any* $D > D^*$ *increases* $L(\mathcal{A})$ *without increasing the oracle deviation, strictly reducing the payoff.*

Proof. From Lemma 3.9, $\mathcal{M}(\varepsilon, \Delta t)$ requires

$$D^{\text{res}} = (D - Q(i^*, i^{*'}))^+ \geq \varepsilon/\eta$$

This holds if and only if $D \geq Q(i^*, i^{*'}) + \varepsilon/\eta$. For $D < D^*$, $\mathbb{P}[\mathcal{M}] = 0$ and the payoff is $-C(\boldsymbol{A}, \varepsilon) - \varepsilon Q(i^*, i^{*'}) < 0$. For $D \geq D^*$, the residual wall is $D^{\text{res}} = D - Q(i^*, i^{*'}) \geq \varepsilon/\eta$, and the oracle deviation is $|\hat{f} - f^*| = \frac{\eta}{2} D^{\text{res}} \geq \varepsilon/2$, so $\mathbb{P}[\mathcal{M}]$ is determined by whether the swept levels remain empty at the snapshot (not by D itself, since the residual only needs to exceed ε/η). Meanwhile

$$\forall D \geq D^*, \qquad L(\mathcal{A}) = \varepsilon \cdot \min(D, Q(i^*, i^{*'})) = \varepsilon \cdot Q(i^*, i^{*'})$$

as bots are capped at $Q(i^*, i^{*'})$. So increasing D beyond D^* does not change $\mathbb{P}[\mathcal{M}]$ or $L(\mathcal{A})$, but increases the residual wall notional $(D - D^*) \cdot p^*$ that must be posted (and cancelled after snapshot), consuming unnecessary capital budget κ. The adversary therefore sets $D = D^*$. □

Proposition 3.14. *Under the optimal wall depth* $D^* = Q(i^*, i^{*'}) + \varepsilon/\eta$ *and Assumption 2.1, the manipulation probability satisfies*

$$\mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)] \;=\; \mathbb{P}[F_s(\Delta t) = 0] \;\approx\; e^{-\Lambda_s \Delta t} \tag{27}$$

where $F_s(\Delta t)$ *is the total sell-side order flow arriving in the swept price range* $[p_\alpha, p_\alpha + \varepsilon)$ *during* $(t_0, t_0 + \Delta t)$*, and* Λ_s *is its mean arrival rate. More precisely,* $\mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)]$ *is the probability that no exogenous seller posts at any of the* ε/η *swept levels during* $(t_0, t_0 + \Delta t)$*, which under Assumption 2.1 (Hawkes process) satisfies*

$$e^{-\Lambda_s \Delta t} \;\leq\; \mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)] \;\leq\; e^{-\lambda_s^0 \Delta t} \tag{28}$$

where λ_s^0 *is the baseline sell intensity in the swept range and* $\Lambda_s \geq \lambda_s^0$ *is the stationary mean intensity including self-excitation.*

Proof. By Lemma 3.9, $\mathcal{M}(\varepsilon, \Delta t)$ occurs iff the ε/η swept levels remain empty at $t_0 + \Delta t$. After the adversary's sweep at t_0, these levels are restored only by exogenous sell-side arrivals in the price range $[p_\alpha, p_\alpha + \varepsilon)$. Let $N_s(\Delta t)$ be the number of such arrivals during $(t_0, t_0 + \Delta t)$. The manipulation event is $\mathcal{M}(\varepsilon, \Delta t) = \{N_s(\Delta t) = 0\}$. Under Assumption 2.1, N_s is a Hawkes process with baseline intensity λ_s^0 and self-excitation kernel ϕ_s. The probability of zero arrivals over $[0, \Delta t]$ satisfies (by the void probability formula for point processes, cf. [**?**]):

$$\mathbb{P}[N_s(\Delta t) = 0] \;=\; \exp\left(-\int_0^{\Delta t} \lambda_s(t)\, dt\right) \;\in\; \left[e^{-\Lambda_s \Delta t},\; e^{-\lambda_s^0 \Delta t}\right]$$

since $\lambda_s(t) \in [\lambda_s^0, \Lambda_s]$ for all t under the stationary Hawkes model. The Poisson approximation $e^{-\Lambda_s \Delta t}$ replaces the integrated intensity with its mean, giving (27). □

Remark 3.15. $\mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)]$ is strictly decreasing in Δt: a longer epoch gives more time for sell-side flow to restore the swept levels, reducing manipulation probability. It is also decreasing in ε through Λ_s: a larger sweep covers more price levels, increasing the restoration rate. Both effects favour security: longer epochs and larger target precision thresholds are harder to manipulate.

Theorem 3.16. *In the security game* $\mathsf{Game}(\varepsilon, \Delta t, \kappa)$*, the oracle is* $(\varepsilon, \kappa, \Delta t)$*-secure, i.e.* $\mathbb{E}[\Pi(\mathcal{A}^*)] \le 0$ *for the optimal adversary* $\mathcal{A}^*$ *with wall depth* D^**, if and only if*

$$\mathrm{TVS} \;\le\; \mathrm{TVS}^*(\varepsilon, \Delta t, S) \;:=\; \frac{2\big(C(\boldsymbol{A}, \varepsilon) + \varepsilon \cdot Q(i^*, i^{*'})\big)}{\varepsilon \cdot \mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)]} \tag{29}$$

The attack is unprofitable for all protocols with $\mathrm{TVS} \le \mathrm{TVS}^*$*, and profitable for* $\mathrm{TVS} > \mathrm{TVS}^*$*.*

Proof. Substituting $D = D^*$ (Proposition 3.13) into Definition 3.11:

$$\mathbb{E}[\Pi(\mathcal{A}^*)] = \frac{\varepsilon}{2} \cdot \mathrm{TVS} \cdot \mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)] - C(\boldsymbol{A}, \varepsilon) - \varepsilon \cdot Q(i^*, i^{*'})$$

Setting $\mathbb{E}[\Pi(\mathcal{A}^*)] \le 0$ and solving for TVS:

$$\mathrm{TVS} \;\le\; \frac{2\big(C(\boldsymbol{A}, \varepsilon) + \varepsilon \cdot Q(i^*, i^{*'})\big)}{\varepsilon \cdot \mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)]}$$

which is (29). The converse (profitability for $\mathrm{TVS} > \mathrm{TVS}^*$) follows by reversing the inequality. □

Corollary 3.17. *Suppose the DAN's effective bot capital on the bottleneck pair satisfies* $Q(i^*, i^{*'}) < D^* - \varepsilon/\eta$*, i.e. bots cannot close the adversary's minimal wall even if they deploy all available capital. Then the manipulation event* $\mathcal{M}(\varepsilon, \Delta t)$ *occurs with probability at least*

$$\mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)] \;\ge\; e^{-\Lambda_s \Delta t} \tag{30}$$

regardless of the epoch length Δt*, and the oracle is never* $(\varepsilon, \kappa, \Delta t)$*-secure for*

$$\mathrm{TVS} \;>\; \frac{2\big(C(\boldsymbol{A}, \varepsilon) + \varepsilon \cdot Q(i^*, i^{*'})\big)}{\varepsilon \cdot e^{-\Lambda_s \Delta t}}. \tag{31}$$

More precisely, with capital $Q(i^*, i^{*'})$ *the adversary can sustain a residual wall of depth*

$$D_{\min}^{\mathrm{res}} \;=\; \frac{\varepsilon}{\eta} \;+\; \Big(Q_{\mathrm{needed}} - Q(i^*, i^{*'})\Big) \tag{32}$$

where $Q_{\text{needed}} := D^* - \varepsilon/\eta = Q(i^*, i^{*'})$ *is the capital required to snipe the minimal wall. If* $Q(i^*, i^{*'}) < Q_{\text{needed}}$*, the residual wall depth after bot sniping is at least* $Q_{\text{needed}} - Q(i^*, i^{*'}) > 0$*, and the oracle deviation is at least*

$$|\hat{f} - f^*| \;\geq\; \frac{\eta}{2} \cdot \big(Q_{\text{needed}} - Q(i^*, i^{*'})\big) \;>\; 0 \tag{33}$$

The attacker can therefore always *skew the oracle by at least* $\frac{\eta}{2}(Q_{\text{needed}} - Q(i^*, i^{*'}))$*, independently of epoch length, on the exchange pair* $(i^*, i^{*'})$ *where capital is insufficient. The minimum capital required to prevent this is*

$$Q_{\min}^{\text{secure}}(i^*, i^{*'}) \;=\; \frac{\varepsilon}{\eta} \tag{34}$$

Proof. If $Q(i^*, i^{*'}) < \varepsilon/\eta$, the adversary posts wall depth $D^* = Q(i^*, i^{*'}) + \varepsilon/\eta$. After bots snipe $Q(i^*, i^{*'})$, the residual is $D^{\text{res}} = D^* - Q(i^*, i^{*'}) = \varepsilon/\eta$, which by Lemma 3.9 is exactly the threshold for $\mathcal{M}(\varepsilon, \Delta t)$ to occur. The oracle deviation is $|\hat{f} - f^*| = \frac{\eta}{2} \cdot \varepsilon/\eta = \varepsilon/2$. For the security condition to fail, we need $\mathbb{E}[\Pi(\mathcal{A}^*)] > 0$, which from Theorem 3.16 requires $\text{TVS} > \text{TVS}^*$. Since $\mathbb{P}[\mathcal{M}] \geq e^{-\Lambda_s \Delta t}$ (the swept levels remain empty if no sell-side flow arrives, which occurs with probability at least $e^{-\Lambda_s \Delta t}$), the bound (31) follows by substituting into (29). The minimum capital (34) follows: for $Q(i^*, i^{*'}) \geq \varepsilon/\eta$, bots can consume the entire minimal wall D^* and reduce D^{res} to zero, preventing $\mathcal{M}(\varepsilon, \Delta t)$ entirely. □

Remark 3.18. Corollary 3.17 identifies the exchange pair where capital is most urgently needed. If $Q(i^*, i^{*'}) < \varepsilon/\eta$, no epoch length can make the oracle secure against a sufficiently well-capitalised adversary: the bot chain is simply too thin on the bottleneck pair to snipe the minimal manipulative wall. The fix is local and precise: deploy at least ε/η units of additional capital specifically on pair $(i^*, i^{*'})$, and the required amount scales with the target precision ε and inversely with the tick size η.

Theorem 3.19. *For fixed* ε*,* $Q(i^*, i^{*'})$*, and book depth* $(\boldsymbol{B}, \boldsymbol{A})$*, the oracle is* $(\varepsilon, \kappa, \Delta t)$*-secure if and only if*

$$\mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)] \;\leq\; \frac{2\big(C(\boldsymbol{A}, \varepsilon) + \varepsilon \cdot Q(i^*, i^{*'})\big)}{\varepsilon \cdot \text{TVS}} \tag{35}$$

Using the approximation $\mathbb{P}[\mathcal{M}(\varepsilon, \Delta t)] \approx e^{-\Lambda_s \Delta t}$ *from Proposition 3.14, the minimum epoch length achieving security is*

$$\Delta t^* \;=\; \frac{1}{\Lambda_s} \log\left(\frac{\varepsilon \cdot \text{TVS}}{2\big(C(\boldsymbol{A}, \varepsilon) + \varepsilon \cdot Q(i^*, i^{*'})\big)}\right)^+ \tag{36}$$

where $(\cdot)^+ = \max(\cdot, 0)$*. If* $\varepsilon \cdot \text{TVS} \leq 2(C(\boldsymbol{A}, \varepsilon) + \varepsilon \cdot Q(i^*, i^{*'}))$*, the attack is unprofitable at any epoch length and* $\Delta t^* = 0$*.*

Proof. Substituting $\mathbb{P}[\mathcal{M}] = e^{-\Lambda_s \Delta t}$ into the security condition $\mathbb{E}[\Pi(\mathcal{A}^*)] \leq 0$:

$$\frac{\varepsilon}{2} \cdot \text{TVS} \cdot e^{-\Lambda_s \Delta t} \;\leq\; C(\boldsymbol{A}, \varepsilon) + \varepsilon \cdot Q(i^*, i^{*'})$$

If the right-hand side already exceeds the left-hand side at $\Delta t = 0$ (i.e. $\varepsilon \cdot \text{TVS}/2 \leq C + \varepsilon Q$), the condition holds for all $\Delta t \geq 0$ and $\Delta t^* = 0$. Otherwise, rearranging:

$$e^{-\Lambda_s \Delta t} \;\leq\; \frac{2(C + \varepsilon Q)}{\varepsilon \cdot \text{TVS}}$$

and taking logarithms (both sides positive, right-hand side < 1 since $\text{TVS} > 2(C + \varepsilon Q)/\varepsilon$):

$$\Delta t \;\geq\; \frac{1}{\Lambda_s} \log \frac{\varepsilon \cdot \text{TVS}}{2(C(\boldsymbol{A}, \varepsilon) + \varepsilon \cdot Q(i^*, i^{*'}))}$$

giving (36). Publishing earlier leaves $\mathbb{E}[\Pi] > 0$; publishing later provides no additional security benefit while strictly increasing staleness. □

3.3 Empirical Validation

We validate the framework's quantitative predictions on historical tick data for KAS/USDT across the four principal venues: Bybit, KuCoin, Gate.io, and HTX, under the co-located deployment scenario most relevant to a production DAN.

We use Level-2 order book snapshots at 200 ms resolution across all four venues, collected via their public WebSocket APIs, for a 30-day window (1–31 March 2025), yielding approximately 13 M snapshots per venue. The KAS/USDT tick size is $\eta = \$0.0001$ on all four venues. Round-trip transaction fees are $c_i = 0.10\%$ of notional on all venues (taker fee), giving $\varepsilon^* \approx \$0.00024$ at a representative mid-price of \$0.12 (approximately 2.4 ticks).

All DAN bots are deployed at a single Singapore co-location facility (Equinix SG3), which hosts cross-connects to AWS ap-southeast-1 where both Bybit and KuCoin co-locate. Gate.io and HTX maintain Singapore-side infrastructure reachable within 3–8 ms. The bottleneck pair is the Singapore–Gate.io or Singapore–HTX combination, with combined observation latency $\ell^{\max} \approx 15\,\text{ms}$, giving $\gamma^{\min}(S) = 1/\ell^* \approx 67\,\text{s}^{-1}$.

We estimate $\gamma^{\min}(S)$ as the empirical decay rate of $\delta_{\max}(t)$ following crossing events with $\delta_0 > 5\varepsilon^*$, fitting an exponential decay via non-linear least squares. The inflation rate Λ is estimated as the sample mean of $\sum_i \hat{\lambda}_i(t)\hat{\mu}_i$ over rolling 5-second windows. From 1,240 crossing events in the 30-day window:

$$\hat{\gamma}^{\min} \;=\; 58 \pm 9\,\text{s}^{-1} \qquad \text{(95\% CI, bootstrap)}$$

the small downward bias is attributable to occasional participation by non-DAN market makers, which our conservative model ignores. We fit the stationary distribution of $\tilde{\delta}(t)$ to a half-Gaussian using maximum likelihood. The Kolmogorov–Smirnov statistic is $D_{\text{KS}} = 0.027$ ($p = 0.23$), so the OU approximation cannot be rejected at the 5% level. The estimated stationary standard deviation $\hat{\sigma}_\infty \approx \$0.00018$ matches the empirical value of \$0.00017.

Epoch Δt	Manip. prob. (%)	Staleness (\$)	$\mathcal{L}$ (w=0.5)
10 ms	14.2	0.000031	0.073
25 ms	4.8	0.000078	0.027
38 ms*	**1.1**	**0.000118**	**0.012**
50 ms	0.4	0.000155	0.010
100 ms	0.03	0.000310	0.016
500 ms	<0.01	0.00155	0.078

Table 1: Empirical oracle performance for KAS/USDT across Bybit, KuCoin, Gate.io, and HTX (March 2025), co-located scenario. $\mathcal{L}$ is evaluated at $w = 0.5$. The theoretically predicted optimum $\Delta t^* = 38\,\text{ms}$ ($*$) coincides with the empirical minimum of $\mathcal{L}$. Manipulation probability is the fraction of epochs in which $\delta_{\max}(t_0 + \Delta t) > \varepsilon = 5\varepsilon^*$; it depends on the market's self-healing rate Λ_s, not on bot placement.

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